# Priyanshu Khandelwal

GitHub LinkedIn

3rd Year Undergraduate
Department of Computer Application

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## Academic Qualifications

| Year | Degree/Certificate | Institute |
|---|---|---|
| 2023 – Present | BCA (AI & ML) | Galgotias University, Greater Noida |
| 2023 | CBSE (XII) | St. Gabriels School, Jabalpur |
| 2020 | CBSE (X) | Delhi Public School, Jabalpur |

## Technical Skills

- **Programming Languages:** Python, SQL
- **AI & ML:** LangChain, LangGraph, RAG, AI Agents, LLM Integration, Context Engineering, Traditional ML, NLP
- **Backend Development:** FastAPI, Flask, REST APIs, JWT Authentication
- **Databases ORM:** PostgreSQL, MySQL, Prisma ORM
- **Cloud DevOps:** Docker, GitHub, Google Cloud Platform (GCP), AWS

## Work Experience

**Backend Engineer – AI Infrastructure — MyPetAlly** *PostgreSQL, Prisma, OpenRouter, Google Cloud Vision*

- Designed & deployed a scalable AI backend **integrating OCR, LLMs & structured context** for pet healthcare insights.
- Built a **Multi-layer conversational pipeline** with memory management (5K tokens), streaming responses, and system guardrails.
- **Implemented** context compression **Reducing token usage by 42%** and optimizing LLM API **Cost Efficiency**.
- **Developed Emergency Detection** and validation logic with **Production-level reliability** and automated test coverage.

## Key Projects

**Multispectral Leaf Image analysis** (Published in **IJEDR**, 2026) Research Paper GitHub (Dec'25–Jul'26)

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|---|---|
| **Objective** | Developed a DL system for plant disease detection & severity estimation using **RGB** & **multispectral Imaging**. |
| **Approach** | • Trained **Model-A (Multispectral CNN)** on 5 spectral bands & vegetation indices for severity estimation.<br>• Trained **Model-B (RGB CNN)** to classify whether a leaf is healthy or diseased from standard RGB images.<br>• **Implemented preprocessing pipeline** including band stacking, normalization, **resizing** (128×128), and **Performance evaluation** using **Accuracy, F1-score, ROC-AUC, & Confusion Matrix**. |
| **Results** | • Demonstrated strong performance in disease detection and severity estimation, highlighting the effectiveness of combining RGB and multispectral deep learning models for precision agriculture. |

**Agentic AI Job Application Automation System** (Independent Project) GitHub (Jan'26–Feb'26)

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|---|---|
| **Objective** | Developed an **LLM-powered automation agent** to streamline job applications by generating personalized emails, sending applications automatically, and tracking application status. |
| **Approach** | • Built a stateful **LangGraph+LangChain** workflow for application retrieval, email generation, delivery, & tracking.<br>• Used **OpenAI GPT** models to generate personalized job application emails from job descriptions.<br>• Connected with **Google Sheets API** & **Gmail SMTP** for automated email delivery with resume attachments.<br>• Implemented automated scheduling using **GitHub Actions**, enabling continuous monitoring of pending applications and preventing duplicate submissions. |
| **Results** | • **Automated the end-to-end job application workflow**, reducing manual effort while maintaining personalized outreach and application tracking. |

**AI Data Analysis Agent** (Independent Project) GitHub (Jan'25–Mar'25)

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|---|---|
| **Objective** | Developed an **AI Agent** capable of processing Multiple file formats & delivering interactive data analysis. |
| **Approach** | • Extracted data from diverse file types using **pandas, docx, fitz,** and **PIL**.<br>• Achieved advanced language understanding & reasoning by implementing the **LLaMA-4 API**.<br>• Generated insights & visual reports by **Analyzing data**, Answering queries & **Handling follow-ups**. |
| **Results** | • Delivered meaningful outputs by Extracting data, Analyzing content & Leveraging LLaMA-4 Maverick. |

## Extra-Curricular Activities

- **Solved 150+ Leetcode Questions** and **participated** in **7+ Hackathons**.
- **Selected** as **Interviewer** for **2 internship candidates** at **MyPetAlly**, **assessing backend systems** and **AI fundamentals**.
- Participated in **Unifest'24**, the **North India's Biggest Fest**.